

MATHEMATICS

Chapter 8: SEQUENCE AND SERIES



SEQUENCE AND SERIES

Top Definitions

1. A Sequence is an ordered list of numbers and has the same meaning as conversational english. A sequence is denoted by $\langle a_n \rangle_{n \geq 1} = a_1, a_2, a_3, \dots, a_n$.
2. The various numbers occurring in a sequence are called its terms.
3. A sequence containing finite number of terms is called a finite sequence. A finite sequence has the last term.
4. A sequence which is not a finite sequence, i.e., containing an infinite number of terms is called an infinite sequence. There is no last term in an infinite sequence.
5. A sequence is said to be an arithmetic progression if every term differs from the preceding term by a constant number. For example, the sequence $a_1, a_2, a_3, \dots, a_n$ is called an arithmetic sequence or AP if $a_{n+1} = a_n + d$.

For all $n \in \mathbb{N}$, where d is a constant called the common difference of AP.

6. A is the arithmetic mean of two numbers a and b if a, A and b forms AP.
7. A sequence is said to be a geometric progression or GP if the ratio of any term to its preceding term is the same throughout. Constant ratio is a common ratio denoted by r .
8. If three numbers are in GP, then the middle term is called the geometric mean of the other two.

Top Concepts

1. A sequence has a definite first member, second member, third member and so on.
2. The n^{th} term $\langle a_n \rangle$ is called the general term of the sequence
3. Fibonacci sequence $1, 1, 2, 3, 5, 8, \dots$ is generated by a recurrence relation given by

$$a_1 = a_2 = 1.$$

$$a_3 = a_1 + a_2 \dots\dots$$

$$a_n = a_{n-2} + a_{n-1}, n > 2.$$

4. A sequence is a function with a domain, the set of natural numbers or any of its subsets of the type $\{1, 2, 3, \dots k\}$.
5. If the number of terms is three with a common difference 'd', then the terms are $a - d$, d and $a + d$.
6. If the number of terms is four with common difference '2d', then the terms are $a - 3d$, $a - d$, $a + d$ and $a + 3d$.
7. If the number of terms is five with a common difference 'd', then the terms are $a - 2d$, $a - d$, a , $a + d$ and $a + 2d$.
8. If the number of terms is six with a common difference '2d', then the terms are $a - 5d$, $a - 3d$, $a - d$, $a + d$, $a + 3d$ and $a + 5d$.
9. The sum of the series is the number obtained by adding the terms.
10. The general form of AP is $a, a + d, a + 2d, \dots, a + (n - 1)d$. The term a is called the **first term** of AP and d is called the **common difference** of AP. The term d can be any real number.
11. If $d > 0$, then AP is increasing, if $d < 0$ then AP is decreasing, and if $d = 0$, then AP is constant.
12. For AP, $a, (a + d), (a + 2d), \dots, (\lambda - 2d), (\lambda - d), \lambda$ with the first term a and common difference d and last term λ and general term is $\lambda - (n - 1)d$.
13. Let a be the first term and d be the common difference of AP with 'm' terms. Then n th term from the end is the $(m - n + 1)^{\text{th}}$ term from the beginning.
14. Properties of AP
 - i. If a constant is added to each term of AP, then the resulting sequence is also AP.
 - ii. If a constant is subtracted from each term of AP, then the resulting sequence is also AP.
 - iii. If each term of AP is multiplied by a constant, then the resulting sequence is also an AP.
 - iv. If each term of AP is divided by a non-zero constant, then the resulting sequence is also AP.
15. The arithmetic mean A of any two numbers a and b is given by $\frac{a+b}{2}$.
16. General form of GP: $a, ar, ar^2, ar^3, \dots$, where a is the first term and r is the constant ratio. The term r can take any non-zero real number.

17. A sequence in GP will remain in GP if each of its terms is multiplied by a non-zero constant.
18. A sequence obtained by multiplying the two GPs term by term results in GP with common ratio the product of the common ratio of the two GPs.
19. The reciprocals of the terms of a given GP form a GP with common ratio $\frac{1}{r}$.
20. If each term of GP is raised to the same power, then the resulting sequence also forms GP.
21. The geometric mean (GM) of any two positive numbers a and b is given by $\sqrt{ab}$.
22. Let A and G be AM and GM of two given positive real numbers a and b , respectively, then $A \geq G$.
Where $A = \frac{a+b}{2}$ and $G = \sqrt{ab}$
23. Let A and G be AM and GM of two given positive real numbers a and b , respectively. Then, the quadratic equation having a and b as its roots is $x^2 - 2Ax + G = 0$.
24. Let A and G be AM and GM of two given positive real numbers a and b , respectively. Then the given numbers are $A \pm \sqrt{A^2 - G}$.
25. If AM and GM are in the ratio $m:n$, then the given numbers are in the ratio $m + \sqrt{m^2 - n^2} : m - \sqrt{m^2 - n^2}$.

Top Formulae

1. The n^{th} term or general term of AP is $a_n = a + (n - 1)d$, where a is the first term and d is the common difference.
2. The general term of AP given its last term is $l - (n - 1)d$.
3. A sequence is AP if and only if its n^{th} term is a linear expression in n , $A_n = Xn + Y$, where X and Y are constants.
4. Let $a, a + d, a + 2d, \dots, a + (n - 1)d$ be AP. Then
 $S_n = \frac{n}{2}[2a + (n - 1)d]$ or $S_n = \frac{n}{2}[a + \ell]$ where $\ell = a + (n - 1)d$
5. A sequence is AP if and only if the sum of its n^{th} term is an expression of the form $Xn^2 + Y$, where X and Y are constants.

6. If the terms of an AP are selected in regular intervals, then the selected terms form an AP.

Let a_n, a_{n+1} and a_{n+2} be the consecutive terms of AP.

Then $2a_{n+1} = a_n + a_{n+2}$.

7. Let $A_1, A_2, A_3, \dots, A_n$ be n numbers between a and b such that $a, A_1, A_2, A_3, \dots, A_n, b$ is AP. The n numbers between a and b are as follows

$$A_1 = a + d = a + \frac{b-a}{n+1}$$

$$A_2 = a + 2d = a + \frac{2(b-a)}{n+1}$$

$$A_3 = a + 3d = a + \frac{3(b-a)}{n+1}$$

$\dots \dots \dots$

$\dots \dots \dots$

$$A_n = a + nd = a + \frac{n(b-a)}{n+1}$$

8. General term of GP is ar^{n-1} , where a is the first term and r is the common ratio.

9. If the number of terms is 3 with the common ratio r , then the selection of terms should be $\frac{a}{r}, a$ and ar .

10. If the number of terms is 4 with the common ratio r , then the selection of terms should be $\frac{a}{r^3}, \frac{a}{r}, a$ and ar^2 .

11. If the number of terms is 5 with the common ratio r , then the selection of terms should be $\frac{a}{r^4}, \frac{a}{r^2}, a, ar$ and ar^2 .

12. Sum to the first n terms of GP $S_n = a + ar + ar^2 + \dots + ar^{n-1}$

(i) If $r = 1$, $S_n = a + a + a + \dots + a$ (n terms) $= na$.

(ii) If $r < 1$, $S_n = \frac{a(1-r^n)}{1-r}$.

(iii) If $r > 1$, $S_n = \frac{a(r^n - 1)}{r - 1}$.

13. Let $G_1, G_2, \dots, G_n$ be n numbers between positive numbers a and b such that $a, G_1, G_2, G_3, \dots, G_n, b$ is GP.

$$\text{Thus } b = br^{n+1}, \quad \text{or} \quad r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$$

$$G_1 = ar = a\left(\frac{b}{a}\right)^{\frac{1}{n+1}}, \quad G_2 = ar^2 = a\left(\frac{b}{a}\right)^{\frac{2}{n+1}}, \quad G_3 = ar^3 = a\left(\frac{b}{a}\right)^{\frac{3}{n+1}}$$

$$G_n = ar^n = a\left(\frac{b}{a}\right)^{\frac{n}{n+1}}$$

14. The sum of infinite GP with the first term a and common ratio r ($-1 < r < 1$) is $S = \frac{a}{1-r}$.

Some Special Series

1. The sum of first n natural numbers is

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

2. Sum of squares of the first n natural numbers

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

3. Sum of cubes of first n natural numbers

$$1^3 + 2^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4} = \left[\frac{n(n+1)}{2}\right]^2$$

4. Sum of powers of 4 of first n natural numbers

$$1^4 + 2^4 + 3^4 + \dots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

5. Consider the series $a_1 + a_2 + a_3 + a_4 + \dots + a_n + \dots$

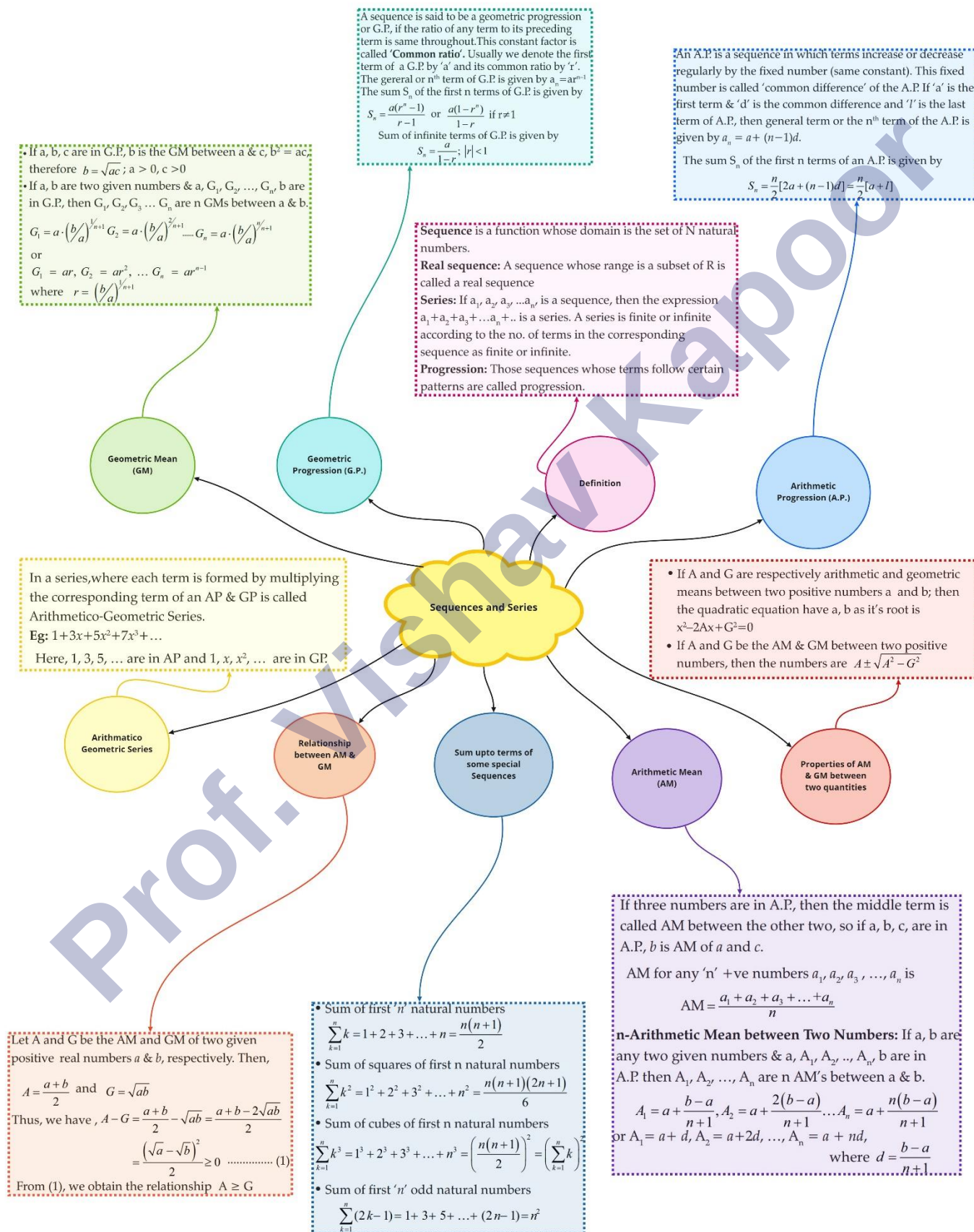
If the differences in $a_2 - a_1, a_3 - a_2, a_4 - a_3, \dots$ are in AP, then the n th term is given by $a_n = an^2 + bn + c$, where a, b and c are constants.

6. Consider the series $a_1 + a_2 + a_3 + a_4 + \dots + a_n + \dots$

If the differences in $a_2 - a_1, a_3 - a_2, a_4 - a_3, \dots$ are in GP then the n th term is given by

$a_n = ar^{n-1} + b + c$, where a , b and c are constants.

Prof. Vishav Kapoor

Class : 11th mathematics
Chapter- 9 : Sequences and Series

Important Questions

Multiple Choice questions-

Question 1. Let T_r be the r th term of an A.P. whose first term is a and the common difference is d . If for some positive integers $m, n, m \neq n, T_m = 1/n$ and $T_n = 1/m$ then $(a-d)$ equals to

- (a) 0
- (b) 1
- (c) $1/mn$
- (d) $1/m + 1/n$

Question 2. The first term of a GP is 1. The sum of the third term and fifth term is 90. The common ratio of GP is

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Question 3. If a is the first term and r is the common ratio then the n th term of GP is

- (a) $(ar)^{n-1}$
- (b) $a \times r^n$
- (c) $a \times r^{n-1}$
- (d) None of these

Question 4. The sum of odd integers from 1 to 2001 is

- (a) 10201
- (b) 102001
- (c) 100201
- (d) 1002001

Question 5. If a, b, c are in AP and x, y, z are in GP then the value of $x^{b-c} \times y^{c-a} \times z^{a-b}$ is

- (a) 0
- (b) 1
- (c) -1

(d) None of these

Question 6. An example of geometric series is

(a) 9, 20, 21, 28

(b) 1, 2, 4, 8

(c) 1, 2, 3, 4

(d) 3, 5, 7, 9

Question 7. Three numbers from an increasing GP of the middle number is doubled, then the new numbers are in AP. The common ratio of the GP is

(a) 2

(b) $\sqrt{3}$

(c) $2 + \sqrt{3}$

(d) $2 - \sqrt{3}$

Question 8. An arithmetic sequence has its 5th term equal to 22 and its 15th term equal to 62. Then its 100th term is equal to

(a) 410

(b) 408

(c) 402

(d) 404

Question 9. Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. If $a < b < c$ and $a + b + c = 3/2$, then the value of a is

(a) $1/2\sqrt{2}$

(b) $1/2\sqrt{3}$

(c) $1/2 - 1/\sqrt{3}$

(d) $1/2 - 1/\sqrt{2}$

Question 10. If the positive numbers a, b, c, d are in A.P., then abc, abd, acd, bcd are

(a) not in A.P. / G.P. / H. P.

(b) in A.P.

(c) in G.P.

(d) in H.P.

Short Questions:

1. Which term of the sequence 27, 24, 21, 18, is zero?
2. The last term of the series 8, 4, 0, is -24 . Find the total number of terms.
3. Seven times the 7th term of a series is equal to eleven times of its 11th term. Find the 18th term of the series.
4. Prove that the sum of $(m + n)^{\text{th}}$ term and $(m - n)^{\text{th}}$ terms of an A.P. is twice of its m^{th} term.
5. Insert three Arithmetic means between 3 and 19.
6. If $-8, A_1, A_2$ are in Arithmetic progression (A.P.), then find the value of A_1, A_2 .
7. A person pays first instalment of Rs. 100 towards his loan. If he increases his instalment every month by Rs. 5, then what will be his 30th instalment?
8. Which term of the sequence $\sqrt{3}, 3, 3\sqrt{3}, \dots$ is 729?
9. How many terms are required in GP. $3, 3^2, 3^3, \dots$, so that their sum would be 120?
10. If A.M. and G.M. of roots of a quadratic equation are 8 and 5 respectively, then find the quadratic equation.

Long Questions:

1. 150 workers were engaged to finish a job in a certain no. of days 4 workers dropped out on the second day, 4 more workers dropped out on the third day and so on. It took 8 more days to finish the work find the no. of days in which the work was completed.
2. Prove that the sum to n terms of the series $11 + 103 + 1005 + \dots$ is $\frac{10}{9}(10^n - 1) + n^2$
3. The ratio of A.M. and G.M. of two positive no. a and b is $m : n$ show that
4. Between 1 and 31, m number have been inserted in such a way that the resulting sequence is an A.P. and the ratio of 7th and $(m-1)^{\text{th}}$ no. is 5:9 find the value of m .
5. The Sum of two no. is 6 times their geometric mean, show that no. are in the ratio $(3 + 3\sqrt{2}) : (3 - 2\sqrt{2})$

Assertion Reason Questions:

1. In each of the following questions, a statement of Assertion is given followed by a corresponding statement of Reason just below it. Of the statements, mark the correct answer as.

Assertion (A) : If the sequence of even natural number is 2, 4, 6, 8, ..., then n^{th} term of the sequence is a_n given by $a_n = 2n$, where $n \in \mathbb{N}$.

Reason (R) : If the sequence of odd natural numbers is 1, 3, 5, 7, ..., then n^{th} term of the

sequence is given by $a_n = 2n - 1$, where $n \in \mathbb{N}$.

- (i) Both assertion and reason are true and reason is the correct explanation of assertion.
 - (ii) Both assertion and reason are true but reason is not the correct explanation of assertion.
 - (iii) Assertion is true but reason is false.
 - (iv) Assertion is false but reason is true.
2. In each of the following questions, a statement of Assertion is given followed by a corresponding statement of Reason just below it. Of the statements, mark the correct answer as.

Assertion (A) : The fourth term of a GP is the square of its second term and the first term is -3 , then its 7th term is equal to 2187.

Reason (R) : Sum of first 10 terms of the AP 6, 8, 10, is equal to 150.

- (i) Both assertion and reason are true and reason is the correct explanation of assertion.
- (ii) Both assertion and reason are true but reason is not the correct explanation of assertion.
- (iii) Assertion is true but reason is false.
- (iv) Assertion is false but reason is true.

Answer Key:

MCQ

- 1. (a) 0
- 2. (c) 3
- 3. (c) $a \times r^{n-1}$
- 4. (d) 1002001
- 5. (b) 1
- 6. (b) 1, 2, 4, 8
- 7. (c) $2 + \sqrt{3}$
- 8. (c) 402
- 9. (d) $\frac{1}{2} - \frac{1}{\sqrt{2}}$
- 10. (d) in H.P.

Short Answer:

1. Here $a = 27$ and $d = T_2 - T_1 = 24 - 27 = -3$.

Let the n^{th} term of series be 0.

$$\therefore T_n = a + (n - 1)d$$

$$\Rightarrow 0 = 27 + (n - 1)(-3)$$

$$\Rightarrow 3n - 3 = 27$$

$$\Rightarrow 3n = 30$$

$$n = 10 \text{ or } 10^{\text{th}} \text{ term.}$$

2. Given series 8, 4, 0, ... (1)

First term = $a = 8$ Common difference $d = 4 - 8 = -4$

$$d = 0 - 4 = -4$$

$\therefore$ The common difference is same the series is in A.P.

last term $l = -24$,

$$l = a + (n - 1)d,$$

$$= -24 = 8 + (n - 1)(-4)$$

$$\Rightarrow -24 - 8 = (n - 1)(-4)$$

$$\Rightarrow -32 = (n - 1)(-4)$$

$$\Rightarrow (n - 1) = \frac{32}{4}$$

$$\Rightarrow n - 1 = 8$$

$$\Rightarrow n = 8 + 1$$

$$\Rightarrow n = 9$$

$\therefore$ Number of terms $n = 9$

3. Let first term = a and common difference = d of A.P.

$\therefore 7^{\text{th}}$ term = $a + 6d$ and 11^{th} term = $a + 10d$.

$\therefore$ According to question,

$$7(a + 6d) = 11 + (a + 10d)$$

$$\Rightarrow 7a + 42d = 11a + 10d$$

$$\Rightarrow 7a - 11a = 10d - 42d$$

$$\Rightarrow -4a = 68d$$

$$\Rightarrow a = -17d$$

$$\text{Hence } 18^{\text{th}} \text{ term} = a + 17d$$

$$= -17d + 17d [\because a = -17d]$$

$$= 0.$$

4. Let the first term = a and common difference = d .

$$T_n = a + (n - 1)d$$

$$T_{m+n} = a + (m + n - 1)d \dots (1)$$

$$T_{m-n} = a + (m - n - 1)d \dots (2)$$

$$T_m = a + (m - 1)d \dots (3)$$

$$T_{m+n} + T_{m-n} = a + (m + n - 1)d + a + (m - n - 1)d$$

$$= 2a + (m + n - 1 + m - n - 1)d$$

$$= 2a + (2m - 2)d$$

$$= 2a + 2(m - 1)d$$

$$= 2[a + (m - 1)d]$$

$$T_{m+n} + T_{m-n} = 2T_m.$$

5. Let three Arithmetic mean be A_1, A_2, A_3 ,
then $3, A_1, A_2, A_3, 19$ are in A.P.
 $\therefore 3 = a, 19 = T_5$, let common difference = d

$$T_5 = a + 4d$$

$$T_5 = a + 4d$$

$$\Rightarrow 19 = 3 + 4d$$

$$\Rightarrow 16 = 4d$$

$$\Rightarrow d = 4$$

$$\text{Hence } A_1 = 3 + 4 = 7, A_2 = 7 + 4 = 11, A_3 = 11 + 4 = 15.$$

6. $-8, A_1, A_2, 9$ are in A.P.

$$\therefore a = -8, T_4 = 9, \text{ let common difference} = d$$

$$T_4 = a + 3d$$

$$\Rightarrow 9 = -8 + 3d$$

$$\Rightarrow 3d = 17$$

$$\Rightarrow d = \frac{17}{3}$$

7. Given : $a = \text{Rs. } 100$, $d = \text{Rs. } 5$, $n = 30$,

$$T_n = a + (n - 1)d$$

Amount of 30th instalment

$$J_{30} = 100 + (30 - 1) \times 5$$

$$= 100 + 29 \times 5 = 100 + 145$$

$$= \text{Rs. } 245.$$

Hence the 30th instalment = Rs. 245.

8.

$$\text{Given : } a = \sqrt{3}, r = \frac{3}{\sqrt{3}} = \sqrt{3},$$

$$\text{Let } n^{\text{th}} \text{ term} = 729$$

$$\therefore T_n = ar^{n-1}$$

$$\Rightarrow 729 = \sqrt{3} \times (\sqrt{3})^{n-1}$$

$$\Rightarrow 729 = (3)^{\frac{1+n-1}{2}}$$

$$\Rightarrow 729 = 3^{\frac{1+n-1}{2}} \Rightarrow 3^6 = 3^{\frac{n}{2}}$$

$$\Rightarrow \frac{n}{2} = 6 \Rightarrow n = 12$$

$$\text{Hence } 12^{\text{th}} \text{ term} = 729.$$

9.

$$\text{Given : } a = 3, r = \frac{9}{3} = 3, S_n = 120,$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$\Rightarrow \frac{3(3^n - 1)}{3 - 1} = 120$$

$$\Rightarrow 3^n - 1 = \frac{120 \times 2}{3}$$

$$\Rightarrow 3^n - 1 = 80$$

$$\Rightarrow 3^n = 81$$

$$\Rightarrow 3^n = 3^4$$

$$\Rightarrow n = 4$$

10. Let the roots of equation be α and β

$$\text{A.M. of roots} = \frac{\alpha + \beta}{2} = 8$$

$$\Rightarrow \alpha + \beta = 16$$

$$\text{G.M. of roots} = \sqrt{\alpha\beta} = 5$$

$$\Rightarrow \alpha\beta = 25$$

If α, β are the roots of equation

$$\text{then, } x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$\Rightarrow x^2 - 16x + 25 = 0.$$

Long Answer:

1. $a = 150, d = -4$

$$S_n = \frac{n}{2} [2 \times 150 + (n-1)(-4)]$$

If total works who would have worked all n days $150(n-8)$

$$A \text{ T Q } \frac{n}{2} [300 + (n-1)(-4)] = 150(n-8)$$

$$n = 25$$

2.

$$S_n = 11 + 103 + 1005 + \dots + n \text{ terms}$$

$$S_n = (10+1) + (102 + 3) + (103 + 5) + \dots + 10n + (2n-1)$$

$$S_n = \frac{10(10^n - 1)}{10 - 1} + \frac{n}{2} (1 + 2n - 1)$$

$$= \frac{10}{9} (10^n - 1) + n^2$$

3.

$$\frac{a+b}{\frac{2}{\sqrt{ab}}} = \frac{m}{n}$$

$$\frac{a+b}{2\sqrt{ab}} = \frac{m}{n}$$

by C and D

$$\frac{a+b+2\sqrt{ab}}{a+b-2\sqrt{ab}} = \frac{m+n}{m-n}$$

$$\frac{(\sqrt{a} + \sqrt{b})^2}{(\sqrt{a} - \sqrt{b})^2} = \frac{m+n}{m-n}$$

$$\frac{\sqrt{a} + \sqrt{b}}{\sqrt{a} - \sqrt{b}} = \frac{\sqrt{m+n}}{\sqrt{m-n}}$$

by C and D

$$\frac{\sqrt{a}}{\sqrt{b}} = \frac{\sqrt{m+n} + \sqrt{m-n}}{\sqrt{m+n} - \sqrt{m-n}}$$

Sq both side

$$\frac{a}{b} = \frac{m + \cancel{m} + m - \cancel{m} + 2\sqrt{m^2 - n^2}}{m + \cancel{m} + m - \cancel{m} - 2\sqrt{m^2 - n^2}}$$

$$\frac{a}{b} = \frac{m + \sqrt{m^2 - n^2}}{m - \sqrt{m^2 - n^2}}$$

4.

$1, A_1, A_2, A_3, \dots, A_m, 31$ are in AP

$$a = 1$$

$$a_n = 31$$

$$a_{m+2} = 314$$

$$a_n = a + (n-1)d$$

$$31 = a + (m+2-1)d$$

$$d = \frac{30}{m+1}$$

$$\frac{A_7}{A_{m-1}} = \frac{5}{9} \quad (\text{Given})$$

$$\frac{1 + 7\left(\frac{30}{m+1}\right)}{1 + (m-1)\left(\frac{30}{m+1}\right)} = \frac{5}{9}$$

$$m = 1$$

5.

$$a + b = 6\sqrt{ab}$$

$$\frac{a+b}{2\sqrt{ab}} = \frac{3}{1}$$

by C and D

$$\frac{a+b+2\sqrt{ab}}{a+b-2\sqrt{ab}} = \frac{3+1}{3-1}$$

$$\frac{(\sqrt{a} + \sqrt{b})^2}{(\sqrt{a} - \sqrt{b})^2} = \frac{2}{1}$$

$$\frac{\sqrt{a} + \sqrt{b}}{\sqrt{a} - \sqrt{b}} = \frac{\sqrt{2}}{1}$$

again by C and D

$$\frac{\sqrt{a} + \sqrt{b} + \sqrt{a} - \sqrt{b}}{\sqrt{a} + \sqrt{b} - \sqrt{a} - \sqrt{b}} = \frac{\sqrt{2} + 1}{\sqrt{2} - 1}$$

$$\frac{2\sqrt{a}}{2\sqrt{b}} = \frac{\sqrt{2} + 1}{\sqrt{2} - 1}$$

$$\frac{a}{b} = \frac{(\sqrt{2} + 1)^2}{(\sqrt{2} - 1)^2} \text{ (on squaring both side)}$$

$$\frac{a}{b} = \frac{2 + 1 + 2\sqrt{2}}{2 + 1 - 2\sqrt{2}}$$

$$\frac{a}{b} = \frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}}$$

$$a : b = (3 + 2\sqrt{2}) : (3 - 2\sqrt{2})$$

Assertion Reason Answer:

- (ii) Both assertion and reason are true but reason is not the correct explanation of assertion.
- (iv) Assertion is false but reason is true.